

Zahra Bashir

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Education

- **University of Alberta** 2021 – 2024
M.Sc. Thesis-based in Computing Science, advised by Prof. Levi Lelis
GPA: 3.8/4.0
Research Topics: Neurosymbolic AI, Program Synthesis, Reinforcement Learning
Edmonton, AB, Canada
- **Iran University of Science and Technology** 2016 – 2020
B.Sc. in Computer Engineering, advised by Prof. Sauleh Eetemadi
GPA: 3.82/4, **Graduated with Honors*
Research Topics: Bandits, Natural Language Processing
Tehran, Iran

Publications

- *Plastic Programming Languages: Learning Neuro-Augmented Domain-Specific Languages*
– **Zahra Bashir**, David Aleixo, Kevin Ellis, Levi Lelis (To be submitted, Abstract)
- *SEGClobber - A Linear Clobber Solver*
– Taylor Folkerson, **Zahra Bashir**, F. Tavakoli, M. Muller (International Computer Games Association, Aug 2025)
- *Revisiting the Assessment of Programmatic Policy Interpretability: Insights from Human Evaluation*
– **Zahra Bashir**, Michael Bowling, Levi Lelis (Under Review)
- *Assessing the Interpretability of Programmatic Policies using Large Language Models*
– **Zahra Bashir**, M. Bowling, L. Lelis (Reinforcement Learning Conference (RLC) InterPol Workshop, June 2024)

Research Experience

- **Research Intern**, University of British Columbia/Vector Institute, Sep 2025 – Dec 2025
– Working on developing policies for physical strategy learning in a few-shot setting.
– Implemented **Bayesian inference** over a space of policies, using priors and likelihoods from demonstrations to guide few-shot learning.
Advisors: Kelsey Allen, Tom Silver
- **Research Intern**, Cornell University, *Remote* May 2024 – May 2025
– Developed a neurosymbolic language augmentation framework combining symbolic search with learned neural components.
– Applied it to **RL** and **Gaussian-process-based time-series forecasting** by adding neural primitives to the program library.
Advisor: Kevin Ellis
- **AI Researcher**, University of Alberta / Amii May 2024 – Aug 2025
– Worked on two research papers: 1) Extended study on interpretability assessment using human evaluation, and 2) Learning Neural Languages with Tokenized Representation.
Advisor: Levi Lelis
- **Graduate Research Assistant**, University of Alberta Jan 2023 – Apr 2024
– Worked on assessing the interpretability of programmatic policies.
– Enhanced programmatic policies for playing MicroRTS through combined tree-search and NN techniques.
Advisor: Levi Lelis
- **CS Research Mentorship Program (CSRMP) Scholar**, Google Sep 2022 – Feb 2023
– Developed research skills and collaborated with Google research teams under the guidance of a Google AI Resident.
- **Research Assistant**, University of Alberta May 2021 – Dec 2022
– Studied MQM-based differentially private generative models for **time-series synthesis**
Advisor: Nidhi Hegde
- **Machine Learning Intern**, Iran University of Science and Technology Jun 2019 – Oct 2019
– Developed an image captioning system using “Show, Attend, and Tell” model, tailored to handle Persian grammar complexities.
Advisor: Dr Mozayyani

Work Experience

- **Machine Learning Intern**, Alberta Machine Intelligence Institute (Amii) May 2022 – Sep 2022
– Supported the industry team by advising companies and clients on applying ML to real-world problems.
– Reviewed and tested MLOps tools (Snowflake, dbt, Amazon S3, Metaflow) on benchmark tasks. *Supervisor: David Chan*
- **ML Project Validator**, Alberta Machine Intelligence Institute (Amii) Feb 22 – Apr 22, Nov 23 – Feb 24
– Conducted literature search on machine learning techniques that could be applied as ML solutions for specific client cases.
Topics included: RAG, LLMs, Recommender Systems.
- **Data scientist**, Sharif Plus (University-based Startup) Jul 2020 – Dec 2020
– Developed a GAN-based approach (LSTM/CNN) for a prediction task on a time-series dataset and used reinforcement learning for online hyperparameter optimization.
- **Developer and Technical Manager**, Chillin Wars AI Contest Sep 2018 - Feb 2019
– Led the technical team for Iran University of Science and Technology’s **ChillinWars AI contest**, an annual well-known AI-programming competition.
– Worked as a full-stack developer of the **Junior Game** for this competition, building on its exclusive framework.
- **Back-end Developer**, D&C (Ravandiyar) Jun 2018 – Sep 2018
– Developed blockchain wallet applications using Django REST.

Awards and Honors

- Admitted to **Princeton University** for a **PhD** in Electrical and Computer Engineering for Fall 2025 (Unable to attend due to U.S. travel ban on Iranian citizens); Advisor: Prof Tom Silver Feb 2025
- Received Horizon Program Fellowship, awarded by Princeton University. March 2025
- Admitted to the DLRL2024 Summer School Organized by CIFAR/Vector April 2024
- Consistently ranked within the **top 3** students out of 100 throughout the undergraduate CE program. 2016-2020
- Member of the ACM ICPC team of the Computer Engineering department. 2017
- Ranked within the **top 0.2%** of the candidates in the "Iranian University Entrance Exam" for bachelor's degree. 2016

Research Interests

- Machine Learning
- Reinforcement Learning
- Neurosymbolic AI
- Planning
- Bayesian Statistics
- Explainability/Interpretability

Teaching Experience

- **Search and Planning in AI & Foundation of Computation II** Jan 2021 – April 2024
 - Facilitated collaborative lab sessions to address coding challenges and conceptual issues for these two courses.
 - Marked assignments and exams.
- **Teaching Assistant for 9 Entry/Medium Level Courses** Sep 2017 – May 2020
 - List of courses: Theory of Languages & Automata, Computational Intelligence, Artificial Intelligence, Discrete Math, Signal & Systems, Software Engineering, System Analysis, Programming Basics.
 - Held workshops and teaching sessions, conducted labs, designed and marked assignments (e.g., course link).

Selected Projects

- **Combinatorial Game Theory-informed Strong Clobber 1-d Solver** (Github Link)
 - Studied and implemented various CGT techniques to create the strongest solver possible in speed and correctness.
 - Verified some hypotheses about game values, and found some interesting game values.
- **Private Time-Series Dataset Generation**
 - Studied privacy in time-series with the goal of releasing a private query for a time-series dataset using two approaches: TimeGAN (model-based approach) and MQM (data-driven approach).
- **Adversarial Attacks on Language Models Using Text-GAN** (GitHub Link)
 - Developed GAN-based adversarial attacks in reduced-dimensional space on language models, achieving 89.95% test accuracy
- **Generating Differential Private Synthetic Data** (GitHub Link)
 - Implemented 3 differentially private GANs (PATE, DP, SPRINT), and applied the PATE method on CGAN.
 - Achieved the highest precision (0.93%) and accuracy (0.83%) for the PATE-ACGAN model compared to SOTA models.
- **NRLP, Propaganda Detection using Multi-Armed Bandit Algorithms** BSc final Project
 - Detection of Propaganda Techniques in News Articles (GitHub Link)
 - Used Thompson Sampling for propaganda-field detection. (Presentation Link)
- **Selected Course Projects (2016-2020)**
 - Designed a noise-robust image detection model using **Hopfield Network**. (GitHub Link)
 - Solved the Inverted Pendulum problem using **Fuzzy Logic** and RL. (GitHub Link)
 - Applied **Genetic Algorithm** to find polynomial equation roots. (GitHub Link)
 - Implemented **Kohonen's** Self-Organizing Feature Map (SOFM) to map 3D data into 2D space (GitHub Link)
 - Additional Computer Vision and NLP projects: (Smile Detection), (political vision detection), (Face Recognition)

Invited Talks and Presentations

- **Reinforcement Learning Conference (RLC) InterPol Workshop - Amherst, MA** Aug 2024
 - * Oral presentation of: "Assessing the Interpretability of Programmatic Policies using LLMs". [SLIDES, POSTER]
- **RLAI Summit 2024 - University of Alberta/Amii** Aug 2024
 - * Presentation on "Learning Neurosymbolic Languages to Solve Reinforcement Learning Problems". [SLIDES]
- **Upper bound 2023 - Neurosymbolic Programming Workshop** May 2023
 - * Exploring the idea of "Using LLMs to Understand Programmatic Policies".

Academic Service

- Reviewer, Association for the Advancement of Artificial Intelligence (AAAI) Conference, Aug 2025
- Reviewer, Conference on Robot Learning (CoRL), Learning Effective Abstractions for Planning (LEAP) Workshop, Sep 2025

Volunteer Experience

- Member of Computer Engineering Scientific Association (CESA) Sep 2018 - Sep 2019
- Member of Iranian Students Association University of Alberta (ISAU) Nov 2021 - Nov 2022